

# **Introduction to cryptography**

# **Learning objectives**

- **Define cryptography**
- **Explain why it is important**
- **Explore how it is used to protect information in cybersecurity**

# **What is cryptography?**

- **The science of securing data**
- **The use of the process of encryption**
- **The protection of personal messages, bank details and passwords**

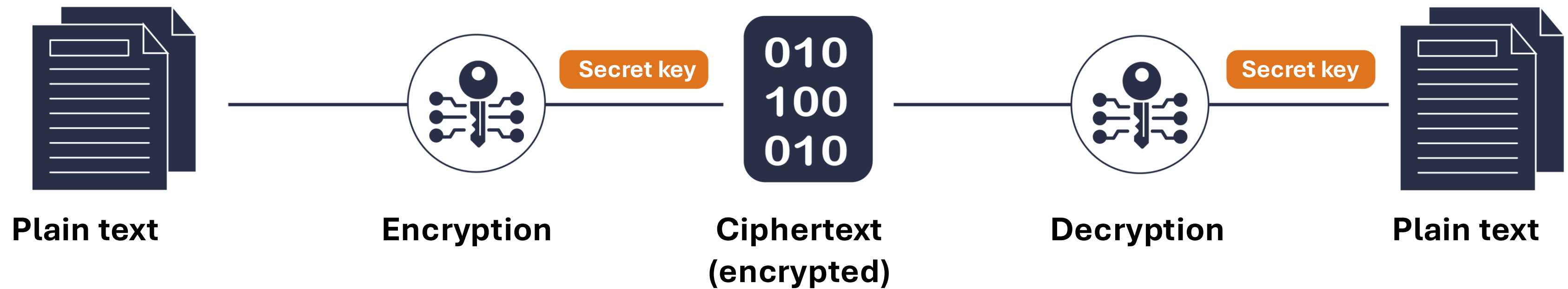
# **Why cryptography matters**

- **Protects privacy and sensitive data**
- **Prevents tampering or unauthorised access**
- **Builds trust in digital systems**

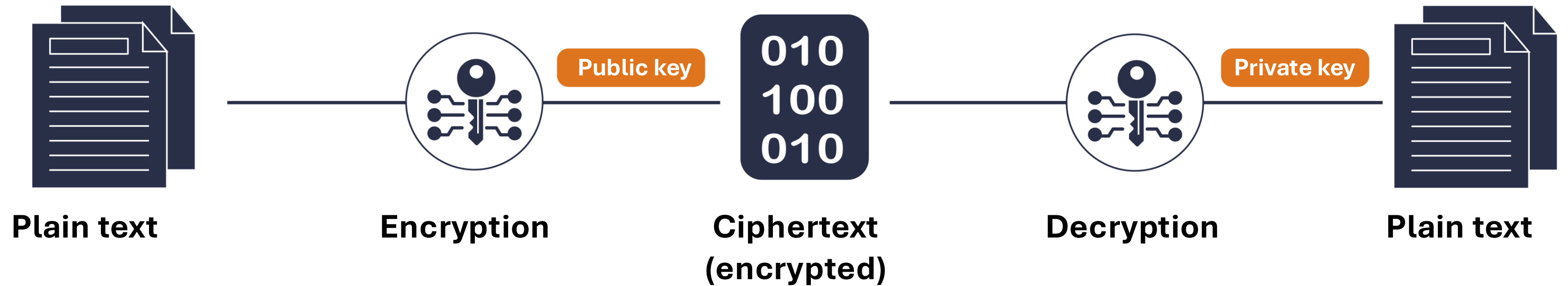
# **Key terms in cryptography**

- **Encryption: changing readable data into scrambled form**
- **Decryption: returning scrambled data to original form**
- **Key: a secret password or number used in both processes**
- **Cipher: the method used to encrypt or decrypt**

# Symmetric cryptography



# Asymmetric cryptography



# **Cryptography in the real world**

- **Secure web browsing**
- **Digital signatures and authentication**
- **Banking and online payments**
- **Messaging apps**

# **Strengths and challenges**

## **Strengths**

- **Strong privacy and trust**
- **Supports digital identity**

## **Challenges**

- **Key management is difficult**
- **Some systems are vulnerable to weak algorithms or human error**

# Summary

- **Cryptography is the science of protecting information**
- **Encryption**
- **Symmetric and asymmetric cryptography**
- **Real world uses**
- **Importance of protecting keys**

# Next

- **Keys management and how symmetric encryption works**
- **Advanced Encryption Standard (AES) and other key example algorithms**